5 Epistemic transparency and epistemic opacity

In the preceding chapters, I defended the central two-dimensionalist claim that every linguistic expression which is a candidate for having an extension has a primary intension: In chapter 3 I showed that this thesis can be upheld even with respect to proper names and natural kind terms, which are often believed to have no a priori implications. In chapter 4, I discussed more general arguments in support of primary intensions and argued that they play important roles in the epistemic determination of one's subject matter and in communication. Now it is surely a crucial step in a defense of conceptual analysis to establish the thesis that words and sentences are indeed associated with primary intensions. But still, even if that much has been achieved, the usefulness of conceptual analysis as a method in philosophy (and elsewhere) seems highly dependent on issues concerning epistemic transparency and epistemic opacity, which will be discussed in this chapter.

According to the analyses I gave of names, natural kind terms and any terms which are deferentially used, these terms are not semantically neutral, i.e. their primary intensions differ from their secondary intensions. This entails that they are epistemically opaque: the (metaphysically) necessary and sufficient conditions for membership in the respective categories are not accessible a priori. Since the notions of epistemic transparency and semantic neutrality will be at center stage in this chapter, it will be useful to explicitly define them:

(SN) An expression is **semantically neutral** iff its primary intension coincides with its secondary intension.

(ET) An expression is **epistemically transparent** iff its secondary intension is accessible a priori.

Epistemic opacity should then be considered as the negation of epistemic transparency, i.e. an expression is epistemically opaque if and only if its

secondary intension is *not* accessible a priori. The relation between (SN) and (ET) will be spelt out in detail in 5.2.

Epistemically opaque terms are at least potentially problematic for the prospects of a conceptual analysis. Since their secondary intensions and thus their metaphysical (or secondary) application conditions are not accessible via the possession of the relevant concepts, it seems questionable if anything of philosophical interest can be learned from them. Another, related problem is that there is often intersubjective variation in the primary intensions of epistemically opaque terms, as we saw in chapters 3 and 4. As I pointed out at the end of chapter 3, such intersubjective variation is a genuine problem for conceptual analysis, understood as a public enterprise. The question of what can be learned from (an analysis of) epistemically opaque terms will be addressed in section 5.3. But first, in 5.1, I will discuss and ultimately reject arguments from Ruth Millikan, Hilary Putnam, and Hilary Kornblith, which are supposed to show that epistemic opacity is ubiquitous. Then, in 5.2, I will deal with the question of whether and how one can determine which kinds of terms are epistemically opaque and which are epistemically transparent.

5.1 Arguments for ubiquitous opacity

Most philosophers agree that the arguments from Kripke, Putnam and others have shown that proper names, natural kind terms and indexical expressions are epistemically opaque. However, Putnam's claim that the same kinds of considerations apply to "the great majority of all nouns, and to other parts of speech as well" (Putnam 1975, 242) has met with considerably less approval. I nevertheless do not think that one can safely consider epistemic opacity as a phenomenon which is clearly limited in scope. For even though Putnam's own attempts to back up his claim may not be entirely convincing,[96] other authors, such as Tyler Burge, Millikan

[96] I will say more about this issue in 5.1.2.

and Kornblith, have brought forth different arguments to show that many other kinds of expressions exhibit opacity as well. I should note that these philosophers typically deny that the relevant expressions have Fregean senses at all, and would therefore most likely reject the idea that they have primary intensions. But since the existence of primary intensions has been defended in chapters 3 and 4, I will leave these issues aside here and focus on questions concerning epistemic opacity.

5.1.1 Millikan

Millikan is probably the most radical of the authors just mentioned. She herself notes that her view goes beyond that of Putnam, Burge, and others (cf. Millikan 2005, 67). Millikan thinks that 'empirical concepts', including even concepts such as 'square', are generally not associated with Fregean senses (cf. Millikan 2010, 50f.; cf. also Millikan 2005, 66). In her view, linguistic expressions are associated with conceptions. But these are very different from senses. A conception corresponds roughly to a collection of (empirical) ways of recognizing the concept's referent. Conceptions neither determine the reference, nor are they generally shared among speakers (cf. Millikan 2005, 67–71). For Millikan, possession of a concept is essentially constituted by a subject's ability to distinguish the referent. This ability need not be infallible; in fact, it can even be based on a mechanism which is only reliable in the subject's environment. Consequently, for two persons to share a concept, and to be able to communicate successfully, they only need to be able to distinguish the same referent. As Millikan puts it, all that is required is agreement in judgment, not in the method(s) of recognizing the referent (cf. Millikan 2010, 56f.).

As a general model of what it takes to possess and to share a concept, Millikan's account seems highly counterintuitive. It may have some plausibility for some kinds of concepts, such as natural kind concepts; but not for concepts such as 'square'. Take for instance a subject who lives in an environment where most (or maybe all) square things are red. Due to her excellent color vision, she can recognize these reliably. By applying the

term ‘square’ to all and only red things, she may even agree in judgments with others and thus, on Millikan’s account, communicate with them. But this person may be unable to recognize shape properties; she may have no idea what shapes are or, somewhat less extremely, she may not know this particular kind of shape. I think that in such a case, we would have no inclination to say that this person possesses the concept ‘square’. (It is a lot more plausible that she possesses the concept ‘red’.)

But be that as it may, a two-dimensionalist is not committed to any particular thesis about the individuation of concepts, i.e. about what it takes for two persons to share a concept (as I pointed out in chapter 3). What she is committed to, though, is that the properties a subject associates with a concept determine the reference with respect to any world considered as actual. As I said above, I will not discuss this thesis again in this chapter. Here, I will try to establish two theses which are also important for a vindication of the method of conceptual analysis: Namely, that at least in the case of some kinds of terms, these associated properties a) are shared between speakers; b) determine the term’s secondary intension independently of features of the actual world.

Both of these claims are rejected by Millikan. However, as she herself concedes sometimes defining features are “passed on explicitly from generation to generation” (Millikan 2005, 71), for instance in the case of terms for geometrical figures. Now if these features determine the terms’ secondary intensions, as is indeed very plausible, then why should one not say that such terms are epistemically transparent terms whose primary intensions are furthermore shared among the speakers of a linguistic community? I think that as it stands, Millikan’s view is not very plausible. At least some of the concepts to which her account is supposed to apply are by all appearances epistemically transparent. If one wants to hold that epistemic opacity does not just concern natural kind terms, names and

indexicals, then one should give some positive arguments to back up this claim.[97]

5.1.2 Putnam

An obvious way to support the idea that epistemic opacity does not only concern names, natural kind terms and indexicals is to try and apply the same kinds of arguments which have been used to show that these terms are opaque to other kinds of expressions. Putnam thinks that 'Twin Earth style' considerations are applicable to many other kinds of terms, such as artifact terms. Recall the following key insights of his about natural kind terms: Firstly, these terms pick out the nature of the corresponding kind rigidly. Secondly, we do not have a priori insight into the essential properties of such kinds and thus into the (secondary) application conditions of the terms. Furthermore, Putnam argues that typically nothing we associate with natural kind terms is a priori.[98] Now, Putnam's remarks about artifact terms exactly parallel those about natural kind terms, as in his famous discussion of the term 'pencil': According to Putnam, it could turn out that pencils are not artifacts, but living organisms:

> We cut them open and examine them under the electron microscope, and we see the almost invisible tracery of nerves and other organs. We spy upon them, and we see them spawn, and we see the offspring grow into full-grown pencils. (Putnam 1975, 242)

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[97] For the sake of fairness, I should add that Millikan does make a positive case for her view, which is mainly based on the Burgean idea that all or nearly all kinds of concepts can be possessed despite the subject's incomplete understanding. Since Kornblith's argument, which will be addressed below, is based on this idea as well, I will postpone a discussion of this issue.

[98] In this respect, Putnam's view seems to be stricter than that of Kripke who holds that at least in some cases, possession of a natural kind concept requires a priori knowledge of a reference-fixing description (cf. the discussion of 'light' and 'heat' in Kripke 1980, 131).

The hypothetical scenario suggests that not even the fact that pencils are artifacts can be known a priori. From this, Putnam infers that the term 'pencil' has no a priori associations. Nevertheless, he does not think that it is metaphysically possible that pencils are organisms. In his view, if the things we call 'pencils' are in fact, as is very likely, artifacts, then they are necessarily artifacts. This is because "[w]hen we use the term 'pencil', we intend to refer to whatever has the same *nature* as the normal examples of local pencils in the actual world." (Putnam 1975, 243) The upshot of Putnam's considerations is that there is no difference in the semantics of natural kind terms and artifact terms. Both kinds of terms refer rigidly to a hidden nature which is not a priori accessible.

Stephen Schwartz was one of the first to criticize Putnam's account of artifact terms (cf. Schwartz 1977, 1978, 1980). He conceded that Putnam's 'pencil' example[99] shows that it is epistemically possible that pencils are not artifacts and thus that it is not a priori that they are. But this by no means shows that 'pencil' has no priori associations or that it is not synonymous with any associated description. It only shows that 'being an artifact' is not part of the correct description (cf. Schwartz 1978, 569). Something similar holds for many other artifact terms, such as 'salt shaker', 'lamp' and 'screwdriver'. It is not a priori that these are artifacts. But there are other features which are more plausibly a priori associated, for example those specifying the function and maybe the form of these items.

Admittedly, it is often not easy to spell out such a priori associated characteristics precisely. But this is just an instance of the general problem that it is often difficult to give definitions of the expressions we use.[100] The more relevant question in the current context of epistemic transparency and opacity is whether we can determine the extension of such terms in counterfactual scenarios a priori. This is the case if their secondary intensions do not vary with contingent empirical facts. In my view, the second part of Schwartz's critique of Putnam in which he invokes a

[99] According to Putnam himself, he borrowed the example from Rogers Albritton (cf. Putnam 1975, 242).

[100] I will discuss this issue in some detail in chapter 7.

hypothetical case bears precisely on this issue: Suppose that every pencil in our environment, or maybe even every pencil in the universe, is an artifact. According to Putnam, this would imply that pencils have an 'artifactual nature' and are thereby necessarily artifacts. But this seems implausible, since it should nevertheless be possible that people start to grow pencils (cf. Schwartz 1978, 570). Elsewhere, Schwartz discusses the following case: Suppose, by analogy with Putnam's story about cats, that all actual salt shakers turned out to be spying devices from Mars. If we then started building small cylindrical objects with small holes at the top and used them to dispense salt, these would nevertheless be salt shakers (cf. Schwartz 1983, 479, fn. 6).
Schwartz concludes that, against Putnam, natural kind terms function very differently than artifact terms, which, following Locke, he takes to be 'nominal kind terms'. While the former ones thus refer to a hidden nature which can only be revealed empirically, the latter ones have application conditions which are accessible to a subject via the possession of the relevant concepts.

In my view, Schwartz' critique of Putnam is essentially correct. My case for the claim that typical artifact terms like 'salt shaker' or 'chair' are epistemically transparent will in part be based on considerations similar to the ones made by Schwartz. But before I set forth my own line of argument, let me sketch Kornblith's argument for ubiquitous opacity, which is based on considerations quite different from the ones of Putnam just outlined.

5.1.3 Kornblith

Kornblith is unimpressed by Schwartz' arguments. He insists that the semantics of artifact terms is no different from that of natural kind terms (cf. Kornblith 1980, 2007b). To show this, he appeals to a line of argument which is already present in Putnam's and Kripke's writings and which was developed further by Burge: Kornblith argues that successful reference in

spite of incomplete understanding, which figures in the argument from Ignorance and Error, does not only occur in the case of names and natural kind terms, but also of artifact terms.[101] In his view, this shows that the reference (and the content) of artifact terms is not determined by speaker associations, either. Consequently, the possession of such concepts does not give a subject access to their (secondary) application conditions.

Kornblith believes that he can nevertheless account for the difference in our intuitions about hypothetical scenarios involving natural kind terms such as 'water' and gold' on one side and 'pencil' and 'salt shaker' on the other. In his view, the difference in our verdicts about cases does not indicate a difference in the semantics of these kinds of terms, but rather in our empirical acquaintance with their referents. He thus admits that subjects usually have better insight into the essence of salt shakers and pencils than of many natural kinds, which is reflected in our judgments about hypothetical cases. But as the possibility of incomplete understanding shows, this is not necessarily so. It is therefore not the mere possession of a concept which provides the subject epistemic access to the nature of the denoted category, but rather empirical knowledge. In other words: Artifact terms are epistemically opaque. Accordingly, Kornblith holds that arguments from Ignorance and Error show that there is no (epistemically) relevant semantic difference between natural kind terms and artifact terms.

There is little reason to think that the same line of reasoning cannot be applied to many other kinds of terms as well. As was already argued by Putnam and especially by Burge, incomplete understanding is a ubiquitous phenomenon which concerns many different kinds of terms. In Kornblith's case, it is natural to suspect that he does not take the relevance of his result to be restricted to artifact terms, either. To support this suspicion, let me quote a remark from a paper of his on philosophical methodology: "I see the investigation of knowledge, *and philosophical investigation generally*, on the model of investigations of natural kinds" (Kornblith 1998, 134, my

[101] I already gave two examples of artifact terms where this will plausibly be the case for a great number of speakers in chapter 3, namely 'MRI' and 'transistor'. Kornblith also lists a couple of such terms, for example 'rheostat' and 'buckboard'.

emphasis). Now, although Kornblith does in fact hold that knowledge is a natural kind (cf. e.g. Kornblith 1998), it would be odd to assume that all philosophically relevant terms refer to natural kinds (or artifacts). A more charitable explanation of the remark just cited is that Kornblith believes that his argument concerning the semantics of artifact terms generalizes and can thus be used to establish the ubiquity of epistemic opacity. In its most ambitious form, it would go like this: All kinds of terms can be subject to incomplete understanding. Therefore, reference (and content) is never determined by speaker associations, which implies that there are no epistemically transparent terms. This conclusion could then in turn be used to argue that conceptual analysis is generally useless. In fact, this might be precisely what Kornblith, who is a well-known critic of conceptual analysis, has in mind.

One should note that when a concept is epistemically opaque, this need not imply that it is ineligible for any kind of conceptual analysis; I will say more about this issue in 5.3. Nevertheless, if all terms, or nearly all terms, were opaque, this would be worrisome for conceptual analysis. Furthermore, although it is not completely obvious that successful reference despite incomplete understanding is possible for all terms, I concede that it can concern a great number and variety of terms, including philosophically relevant ones. So I think if one aims to defend conceptual analysis against the objection from incomplete understanding, one has to reject the step from the premise that a subject can refer by using a term which she understands incompletely to the conclusion that that term is epistemically opaque. Prima facie, this seems difficult: In chapter 3, I argued that the argument from Ignorance and Error does not show that such concepts have no primary intensions and proposed to consider them as deferential. But the analysis I gave there showed that the primary intensions of deferential concepts differ from their secondary intensions; they are thus epistemically opaque.

One way in which a proponent of conceptual analysis might want to attack Kornblith's argument is by attempting to show that there has to be a difference in the semantics of natural kind terms and artifact terms, since

there must be speakers with complete understanding of an artifact term. It will be instructive to consider why such an argument fails. After that, I will set forth my own argument, in which I concede that artifact terms, and other kinds of terms as well, can be subject to incomplete understanding to the same extent as, say, natural kind terms.

The idea of Putnam and others is that in the case of natural kind terms, nature does the linguistic labor. When someone introduces such a term, she only needs to be acquainted with an exemplar of a kind and intend to refer to the particular natural kind exemplified. In the case of artifact terms, however, nature will hardly do the linguistic labor. This suggests that when a person has an insufficient grasp of the category denoted by a term, there have to be other members in her linguistic community – the 'experts' – with complete understanding to whom she can defer and who thus do the linguistic labor for her. But in fact, this need not be so, as an example invoked by Kornblith (cf. Kornblith 2007b, 146) which I modified slightly shows: Suppose that archaeologists find a number of artifacts from a long extinct culture which they take to belong to a common kind and introduce a term for that kind of artifact.[102] Such a case, in which there would be no-one in the linguistic community who knows the metaphysical conditions for membership in the relevant kind, seems clearly possible. It is therefore not necessary that there are subjects with complete understanding in the case of artifact terms, either.

Admittedly, the account I just gave of the introduction of the artifact term is a bit too simple. For instance, the ancient artifacts discovered may have had many different functions and thus exemplify more than one kind of artifacts. Therefore, to secure reference to a particular kind of artifact, it will not do to point at some exemplars. So prima facie, there has to be at least one person, namely the one who introduces the term, who knows more about the category denoted by the term after all.

In the case of natural kind terms, there is a very similar problem, which Devitt & Sterelny call the 'qua problem' (cf. Devitt & Sterelny 1999, 90–93): A particular specimen of a natural kind usually exemplifies many

[102] In Kornblith's example, the term introduced refers only to a single artifact.

different kinds; for example a eukaryote, a carnivore, a mammal, a tiger, … Accordingly, it has to be specified to which kind the term is to refer in the case of natural kind terms, too. And therefore, one cannot establish that there is a semantic difference between natural kind terms and artifact terms with an appeal to the problem just mentioned, anyway. Furthermore, this problem only concerns the introduction of a term. But the person who once introduced the term may have long forgotten what she intended the term to refer to or may no longer be alive. And maybe most importantly, even if there is someone whose knowledge or understanding of the term is sufficient to secure reference without deferring to others, the term can still be epistemically opaque for her.
A lesson to be drawn from these considerations is that any semantic difference between natural kind terms and artifact terms which one can hope to find will be a contingent one: No such difference can be derived from the difference in the metaphysical status of their referents alone.

In the following, I will argue that despite the fact that incomplete understanding can concern artifact terms to the same extent that it concerns natural kind terms, there is an epistemically relevant semantic difference between terms like ‘water’ and ‘gold’ on one hand and ‘salt shaker’ and ‘pencil’ on the other. The simple idea I want to defend is that the latter ones, along with many other terms which can be subject to the division of linguistic labor, are epistemically transparent for those with complete understanding of the relevant terms. It is clear that Kornblith’s argument from the possibility of Ignorance and Error does not undermine this claim. In fact, the phrase ‘incomplete understanding’ already suggests that there can also be complete understanding. I noted in chapter 3 that it is hard to say what complete understanding could mean in the case of natural kind terms. In the case of artifact terms, I think it is intuitively more plausible that they have a one-dimensional, i.e. semantically neutral, meaning which just need not be known to every speaker. In the following, I will yield some support to this idea.

Generally speaking, in order to show that a term is epistemically transparent, one has to establish two things: firstly, that it has a primary intension; secondly, that its reference with respect to counterfactual worlds does not vary with contingent features of the actual world. In my view, hypothetical cases of the kind invoked by Schwartz are perfectly suited to show this. More specifically, I will argue in the following that, against Kornblith, the best explanation for the difference in our judgments about hypothetical scenarios involving terms like ‘gold’ and ‘water’ on the one hand and ‘salt shaker’ and ‘pencil’ on the other is that the latter ones, unlike the former, are epistemically transparent.

Consider again Schwartz’ discussion of the term ‘salt shaker’: His considerations suggest that it is epistemically necessary that salt shakers have a specific function (and maybe form). It seems inconceivable that objects which were not designed, never used and never even intended to be used to dispense salt and which are moreover not usable to do so are nevertheless salt shakers. Furthermore, Schwartz’ thought experiment about the spying devices from Mars suggests that such functional properties also reflect the term’s application conditions with respect to counterfactual scenarios: No matter what the things we call ‘salt shakers’ turn out to be, an object with the appropriate function (and maybe form) would still be a salt shaker. Kornblith does not dispute the correctness of these intuitive judgments about hypothetical cases. But he denies that they are a priori; rather, he says, they are empirically shaped. Most of us are closely familiar with salt shakers and know very well what salt shakers are and what they are for. On the other hand, we are often much less familiar with and have much less well-established beliefs about natural kinds. In Kornblith’s view, this is what explains the difference in our judgments about scenarios involving artifacts on the one hand and natural kinds on the other. There is thus no need to appeal to a difference in the semantics of the corresponding terms.

It is anything but clear, however, that empirical (background) beliefs really influence our conceivability judgments, or more specifically our judgments about worlds which are considered as actual or worlds considered as

counterfactual relative to one considered as actual.[103] Consider the following example: Every raven I have seen in my life was black. Moreover, I firmly believe that there are no green ravens. Nevertheless, I can easily imagine green ravens. Maybe Kornblith would object that theoretical beliefs have to be taken into account as well: For instance, I believe that having a specific color is normally not among the properties essential for membership in a species, so my (empirically-based) background belief that there could have been green ravens may explain why I can imagine them. So here is a different example: Every cat I have seen in my life was an animal and I firmly believe that all the other cats in the world which I have not seen yet are animals as well. Even more so, since I trust in Putnam's and Kripke's insights concerning natural kinds, I believe that cats are necessarily animals. Nevertheless, I can easily conceive of a scenario where cats turn out to be robots.[104]
It is thus hard to see how an appeal to a difference in our empirical acquaintance with the referents of these terms could explain the difference in our judgments about hypothetical cases. A much more plausible explanation is that there is a difference in the semantics of these terms: While the secondary application conditions of 'cat' are dependent on contingent features of the actual world, those of 'salt shaker' are not. Since it is very plausible that the same kinds of considerations apply to many other artifact terms such as 'pencil', 'washing machine' or 'screwdriver', I conclude that these kinds of terms are epistemically transparent.

Let me briefly sum up what has been shown in this section: None of the accounts I considered succeeds in establishing the claim that the phenomenon of epistemic opacity extends far beyond names, natural kind terms and indexicals. The most promising argument in support of such a view, which is based on the premise that many other kinds can be subject to the division of linguistic labor as well, does not undermine the idea that these kinds of terms are epistemically transparent for those with complete

[103] Cf. also my discussion of Schroeter's account in the previous chapter.
[104] Cf. Putnam's famous example in Putnam 1962, 660.

understanding. Furthermore, at least in the case of artifact terms such as ‘salt shaker’ or ‘pencil’, our judgments about hypothetical scenarios involving them are most naturally explained by the fact that they are epistemically transparent.
But as I just mentioned, this is only true for subjects with complete, or non-deferential, understanding of these terms. And while I think it is safe to assume that incomplete understanding of terms like ‘salt shaker’ or ‘pencil’ is quite untypical, one could still wonder more generally if and to what extent incomplete understanding presents a problem for conceptual analysis. I will discuss this question briefly at the end of this chapter. In the following, I will have a closer look at how one can generally determine whether an expression is opaque or transparent.

5.2 Revealing opacity

Since, as was just argued, there are epistemically transparent as well as epistemically opaque terms, it would be useful to get a grip on which (types of) expressions fall into which of these categories. The discussion in the previous section may suggest that while indexicals, names and natural kind terms are opaque, artifact terms are transparent. But in fact, the example about the ancient artifacts which I borrowed from Kornblith shows that this need at least not generally be so. Kornblith therefore rightly remarks that one cannot read off the semantics of a term from the metaphysical status of the category it denotes (cf. Kornblith 2007b, 145). But I think this observation cuts both ways. Natural kind terms are not always epistemically opaque, either; a very simple example is ‘H_2O’: It is both a priori and necessary that H_2O consists of two hydrogen atoms and one oxygen atom.[105] In the following, I will therefore propose two alternative ways to determine whether an expression or a type of expression is opaque or transparent.

[105] I am inclined to grant, however, that proper names and indexical expressions are generally opaque: A transparent name would at least be unusual, while indexicals are opaque pretty much by definition.

5.2.1 Revealing opacity via the function of a term

One way to approach the task of revealing opacity is to consider the function of a term. The main functions of proper names, for instance, seem to be firstly to secure reference to a specific individual which is stable even with respect to modal contexts, and secondly, to coordinate the reference with other speakers.[106] Since we usually do not know the essential properties of the individual referred to, it thereby becomes apparent why the secondary application conditions of names are opaque.

Similar considerations apply to natural kind terms. Often, these terms are introduced in the following way: We come across a number of objects, or phenomena, and come to think that these objects are belonging to a common kind, or that the phenomena are caused by a common kind. When we introduce a term for that kind, we need not, and often do not, know what is essential to it. So it is no surprise that many natural kind terms are epistemically opaque.

In chapter 3, I gave an outline of how stable and rigid reference is achieved in the case of names and natural kind terms. The basic idea is that we draw on contingent features of the individual or kind and then rigidify. In some cases, these are not 'objective' features of the referent, but ones which specify its relation to us – this is where centered contents come into the picture. In other words, the associated properties do not, and are not meant to, reflect such a term's secondary application conditions. Rather, secondary intensions depend on the nature of the individual or kind which actually has these properties.

There are many other terms, however, which serve very different purposes. Take artifact terms: Since artifacts are typically produced to fulfill a specific function, it makes sense to introduce terms which simply pick out any object serving, or being intended to serve, that function. The 'nature' of actual exemplars of the 'artifactual kind' is therefore not relevant for the applicability of such terms. In this respect, the term for the ancient artifacts in the example I discussed above is an exception. However, that term is

[106] I already talked about the way this coordination is achieved in 4.2.

introduced in a way which is similar to how I described the introduction of many natural kind terms: People come across a number of objects which they suspect belong to a common kind and introduce a term for that kind, without knowing what exactly unites these objects, i.e. without knowing what makes them samples of a common kind. This is hardly a typical way of introducing an artifact term. Often, artifact terms are introduced even before any artifacts of that kind exist. In these cases, no story akin to the one I told about the introduction of natural kind terms can possibly apply. Moreover, as I indicated above, natural kind terms do not always work the same way, either. Sometimes, for instance, the existence of a natural kind is postulated (or hypothesized) for theoretical reasons; arguably, essential properties of this hypothetical kind will then be a priori associated with the corresponding term, like (maybe) in the case of 'Higgs boson'. In other cases, like with 'H_2O', the expression is supposed to pick out the essence of a familiar natural kind.

I just argued that typical artifact terms such as 'salt shaker', 'pencil' and 'screwdriver' simply pick out objects with a certain function. This may be a rather trivial point, but I think it reveals an influential mistake made by Kornblith and maybe even more clearly by Putnam: They seem to assume that since referential expressions serve to refer to an object or a class of objects, their (secondary) application conditions must be dependent on the nature of actual exemplars of the relevant category. But there is no reason to assume this, and in fact there are a number of reasons to believe that many kinds of terms work quite differently.

To sum up: In this section, I proposed that considering the function(s) of a term can help determine whether it is opaque or transparent. In turn, to determine its function(s), it can be helpful to think about the possible circumstances of a term's introduction. It transpired that if one applies such an approach to natural kind terms and artifact terms, one is led to expect that 'typical' natural kind terms are opaque, while 'typical' artifact terms are transparent, which confirms the verdict from the previous section.

5.2.2 Revealing opacity via considerations about hypothetical cases

The verdict just mentioned about the semantics of terms like ‘salt shaker’ as opposed to that of terms like ‘cat’ was based on judgments about hypothetical scenarios. The difference in the behavior of transparent and opaque terms with respect to such scenarios thus yields a second way to identify them. This approach can be modeled very well in a two-dimensionalist framework. Consider the two-dimensional matrix for ‘water’:

‘water’	$w_@$	w_2	w_3
$w_@$ (WS: H_2O)	H_2O	H_2O	H_2O
w_2 (WS: XYZ)	XYZ	XYZ	XYZ
w_3 (WS: ABC)	ABC	ABC	ABC

(Figure 6)

The diagonal of the matrix, which represents the primary intension, is accessible a priori – via judgments of the form ‘If the world is such and such, then water is …’. So, as should be familiar by now, grasp of a term’s primary intension enables a speaker to determine its extension with respect to any world considered as actual, if provided with complete empirical information about that scenario. Now recall, from the introduction to two-dimensionalism in chapter 2, that the only reason why we sometimes cannot determine the extension of a term with respect to a counterfactual situation is because we lack knowledge of some relevant contingent features of the actual world. If, for instance, we know that the clear,

drinkable, … liquid in our rivers and lakes is H_2O, then we are in a position to know that the substance on Twin Earth would not be water.

If one conjoins these two ideas, one arrives at the view that we are able to access a term's (hypothetical) extension with respect to a counterfactual world relative to any world considered as actual. Here is an example of such a judgment: 'If the clear, drinkable, … liquid in our rivers and lakes is / had turned out to be XYZ, then the substance on Twin Earth would have been water.' In principle, by running through every counterfactual world relative to a specific world considered as actual, we are thus able to access the (hypothetical) secondary intension of an expression relative to any world considered as actual. One thereby arrives at judgments like 'If the clear, drinkable, … liquid in our rivers and lakes is XYZ, then water is necessarily XYZ' or 'If the realizers of the cat role are robots, then cats are necessarily robots' (where the cat role corresponds to the theoretical role associated with 'cat'). Accordingly, it is not just the diagonal of the matrix which can be accessed a priori; the whole matrix is a priori.[107] The only thing about that matrix which we cannot know a priori is in which of the rows we are located, i.e. which of the centered worlds on the left which can be considered as actual is in fact the actual world. Two-dimensionalism thus underpins the kind of judgments necessary to determine whether a given term is epistemically transparent: One hypothetically varies characteristics of the actual world to see whether this affects a term's extension with respect to counterfactual scenarios and thus its secondary intension. If it does, then the term is epistemically opaque, if it does not, it is transparent. This is exactly in line with the way Schwartz established the thesis that there is a difference in the semantics of 'pencil' and 'salt shaker' on the one hand and 'water' and 'cat' on the other: By reference to (judgments about) hypothetical cases, he showed that while the extensions of the latter ones are dependent on the 'nature' of actual exemplars – and

[107] In Chalmers' terminology, the matrix is the 'two-dimensional intension'; formally speaking, the two-dimensional intension is a function from pairs of worlds considered as actual and worlds considered as counterfactual to extensions. In Chalmers 2006, 102f., Chalmers also claims that the two-dimensional intension is accessible a priori.

thus on contingent features of the actual world –, those of the former ones are not.

5.2.3 Can opacity be determined a priori?

The foregoing considerations suggest that it is generally a priori whether a term is transparent or opaque. But this view faces potential counterexamples. Take the case of 'jade', as it is often told in the philosophical literature:[108] People once thought that jade is a natural kind. But then it was discovered that what is called 'jade' are actually two different, though superficially similar minerals, jadeite and nephrite. Although the jade role, i.e. the cluster of superficial characteristics shared by jadeite and nephrite, and those associated with 'jade', is, at least in our environment, only realized by two substances, there could have been many more substances which shared these features. It seems plausible that these substances would be jade, too. If one has doubts about this, one can change the example slightly: Suppose that it had turned out that the jade role is actually realized by a large number of substances and mixtures of substances. In this case we would surely be inclined to call anything 'jade' which satisfies the jade role.

These considerations suggest that 'jade' is (or could have easily been / turned out to be) a purely functional term, picking out anything which realizes the jade role in every possible world. If, however, it had turned out that what we call 'jade' is actually a natural kind, the term would have rather behaved like a typical natural kind term and picked out the same substance with respect to every world.

Although the case of 'jade' is surely unusual, I nevertheless think it is anything but unique. Take for instance 'air': For a long period in human history, air was believed to be an element. One can reasonably assume that if this had turned out to be correct, 'air' would have turned out to refer

[108] This story is most likely not very accurate, so one should take the case to be partly hypothetical.

rigidly to that element. But in fact, air is just a mixture of gases in varying proportions. Plausibly, ‘air’ picks out anything which plays the air role with respect to every possible world – where ‘the air role’ stands for the theoretical role which is a priori associated with ‘air’. Let me illustrate this with a two-dimensional matrix:

‘air’	$w_@$	w_2	w_3
$w_@$ (AS: mixture of gases)	mixture of gases	A	AB
w_2 (AS: A)	A	A	A
w_3 (AS: AB)	AB	AB	AB

(Figure 7)

(Here, ‘AS’ abbreviates ‘airy stuff’, in analogy with Chalmers’ ‘watery stuff’, and stands for whatever plays the air role in the relevant world. In $w_@$, the air role is played by a mixture of gases, in w_2 by the element A, and in w_3 by the molecular compound AB.)

As I just said, the secondary intension of ‘air’ picks out whatever plays the air role in every world considered as counterfactual. The term is thus semantically neutral, i.e. its primary and secondary intension are equivalent. Terms which have this feature are often considered to be epistemically transparent. But as the matrix shows, there is an important difference between ‘air’ and other semantically neutral expressions: If, for

instance, we consider w_2 as actual, we see that relative to this scenario, the secondary intension picks out the element A in every world considered as counterfactual, including those where A fails to play the air role. So relative to w_2 considered as actual, the primary and the secondary intension of 'air' are not equivalent. This mirrors the fact that if the occupant of the air role had turned out to be an element, the term would have behaved like a standard natural kind term. It is thus epistemically contingent that 'air' is semantically neutral. And this seems to imply that it is not a priori that 'air' is epistemically transparent, in violation of the thesis formulated above.

One possible way to react to the examples I just gave is to say that they involve meaning change. When it turned out, for example, that the stuff called 'air' is not an element (or some other natural kind), it really turned out that the term 'air' is empty. But since it would have been impractical not to have a term for the stuff surrounding us, the meaning of 'air' changed, such that it now refers to the mixture of gases.

It is not easy to argue against such a view. But I think that generally, the default assumption should be that the meaning of an expression remains constant. Admittedly, there are pragmatic reasons to have a term for the stuff surrounding us, but these are likewise reasons to associate the term with a primary intension which does not make the existence of air dependent on it being a natural kind. Furthermore, the cases at hand are at least compatible with the idea that the primary and the secondary intension (and the whole matrix) remains constant, in case of which there would be no reason to say that there was a change in meaning. So even if one holds that the primary or the secondary intensions of 'jade' and 'air' changed when the relevant discoveries were made, there is little reason to think that there are no such terms for which it is not a priori whether they are semantically neutral.

On closer inspection, these examples do nevertheless not really show that epistemic transparency is sometimes only empirically evaluable. To demonstrate this, let me first note that they cannot speak against the idea that we are able to access the whole matrix a priori. If anything, they presuppose precisely this ability – in particular the ability to determine the

extension of an expression with respect to a world considered as counterfactual relative to a world considered as actual. So let me have a closer look at what the cases do show, using the example of ‘air’: Recall that the air role is the theoretical role associated with ‘air’. It is thus a priori that air plays the air role. Since the term’s secondary intension coincides with its primary intension, it is also necessary that air plays the air role. However, as we have seen it is not a priori that air necessarily plays the air role: If for instance w_2 had turned out to be actual, it would have been (metaphysically) possible that air does not play this role. So, although in some loose sense one may say that we know the nature of air a priori, we do not know a priori what the nature of air is. For all we know a priori about air, it could also be essentially A, or AB, or ...
Consequently, since we do not have a priori insight into the secondary application conditions of ‘air’, we should say that the term is not epistemically transparent, despite the fact that its primary and secondary intension are equivalent. Accordingly, semantic neutrality does not entail epistemic transparency. It is only those terms whose secondary intensions coincide with their primary intensions with epistemic necessity which are epistemically transparent.
I conclude that terms such as ‘jade’ and ‘air’ are in fact epistemically opaque. Since it is also a priori that they are opaque, they present no counterexample to the thesis that we are able to determine a priori whether a term is transparent or opaque. What we cannot say a priori of them, however, is whether they are semantically neutral. It was for example only the empirical discovery that air is not a natural kind which enabled us to recognize that the secondary intension of ‘air’ coincides with its primary intension. This entails that in some cases, we cannot say a priori whether an a priori associated property is also a metaphysically necessary property of the denoted category.

This result has implications for the question of our epistemic access to the general modal status of a statement. Kripke holds that although we cannot know the truth-values of many necessary statements a priori, we can still

know a priori that whatever truth-value they have, they have it necessarily (cf. Kripke 1980, 109). If one adds to this the claim that we can also say of contingent statements that they are contingent, i.e. either contingently true or contingently false, one arrives at the thesis that we can generally determine the modal status of a statement a priori.[109] Now consider the sentence 'Air plays the air role': From what I argued above, it follows that the sentence is a priori and expresses a necessary truth. But we cannot know a priori that it expresses a necessary truth, nor can we know a priori that if it is true, it is necessarily true. As I already pointed out above, for all we can say a priori, w_2 could be actual, in which case the statement would be contingent.

Further counterexamples to the thesis that the general modal status of a sentence is determinable a priori can easily be generated. Take for instance 'If air exists, then A exists'.[110] That statement is contingently false, but for all we know a priori, it could be necessarily true – for instance in case w_2 turned out to be actual. So the thesis that we can generally determine the modal status of a statement a priori is false. In fact, that thesis is closely related to the claim that we can say a priori whether an expression is semantically neutral, so it should not come as a surprise that they fall together.[111]

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[109] The phrase 'general modal status' was introduced by Albert Casullo, who discusses, and seems inclined to accept, this thesis in Casullo 2003 and Casullo 2010. For an explicit defense of the thesis cf. Horvath 2009.

[110] Since the term 'water' works similarly – cf. below –, one can also use the following, somewhat nicer example: 'If water exists, then oxygen exists.' That sentence is necessary, but if water had turned out to not be a natural kind (or even a different natural kind), it would have been contingent.

[111] A sentence is semantically neutral if it is either epistemically and metaphysically contingent or epistemically and metaphysically necessary. Since it is plausible that we can determine a priori whether a sentence is epistemically contingent, any sentence of which one cannot say a priori whether it is semantically neutral is such that we cannot say a priori whether it is metaphysically contingent or necessary. But this is just to say that its general modal status is not determinable a priori.

There are most likely further terms of which we cannot say a priori whether they are semantically neutral. For example, there are plausibly at least some terms which are in fact natural kind terms, but which could have turned out to be not. Take 'water': It is natural to think that if Lavoisier had found out that the stuff he investigated is just an inhomogeneous mixture of many different substances, he would not have thereby discovered that there is no water. Rather, in this case the term 'water' would have been applicable to anything playing the water role. It is therefore epistemically contingent that the secondary intension of 'water' differs from its primary intension – the term could have turned out to be semantically neutral.
Nevertheless, it would be wrong to think that all of our terms work like this and that therefore, there are no epistemically transparent terms. There is a large number of semantically neutral terms whose neutrality is not dependent on contingent features of the actual world and which are thus epistemically transparent.[112] Schwartz' discussion of terms such as 'pencil' and 'salt shaker' illustrates this nicely. Furthermore, as I pointed out above, it is also plausible for reasons independent of the evaluation of hypothetical scenarios that there are many terms concerning which the 'nature' of actual exemplars is just irrelevant for their applicability. If so, then despite the counterexamples given, there are still many epistemically transparent terms, i.e. semantically neutral terms whose neutrality can be recognized a priori.

So far, I argued that there are both epistemically opaque and epistemically transparent expressions, which raised the question of whether and how we can decide into which of these categories a given term falls. I defended the view that even though it is sometimes an empirical question whether an expression is semantically neutral, one can still tell a priori whether it is transparent. To put it briefly, an expression is epistemically transparent if it

[112] Likewise, there are presumably many natural kind terms such that if it turned out that there is no such natural kind, then the term is empty.

is semantically neutral with epistemic necessity.[113] This leaves open the following two questions, both of which are highly important with respect to the prospects of conceptual analysis: Firstly, on which side of the divide can we expect to find those terms which are (most) relevant for philosophical inquiry? Secondly, what can be learned from opaque terms or, to put it differently, what can result from a conceptual analysis of such terms?

I will just say a few words about the former question, because I do not think it has a definite answer: Bealer claims that most philosophically interesting concepts can be known determinately (cf. Bealer 1998), which roughly corresponds to the view that they are transparent if understood completely.[114] However, I suspect that philosophically interesting expressions form a very inhomogeneous class, presumably even less homogeneous than natural kind terms and artifact terms. So although I suppose that many terms which have been subject to conceptual analysis, such as 'free will', 'knowledge', 'truth' or 'good'[115], are indeed transparent, Bealer's claim that this is true for most philosophically interesting expressions is quite bold and also hard to evaluate. In any case, one cannot reasonably hope that all terms which might be of philosophical interest are transparent. Therefore the question of what can be learned from opaque terms, which will be addressed in the following, is definitely significant for conceptual analysis.

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[113] Let me add the following reservation, though: Intuitions about possible cases and thus also about opacity can sometimes be inconclusive. Some terms are probably borderline or ambiguous between being transparent and being opaque (as Lewis seems to suggests for 'water' (cf. Lewis 1994, 424)). I nevertheless believe that most terms can be conclusively classified as either opaque or transparent.

[114] A similar view is held by Alvin Goldman (cf. Goldman 2007).

[115] The idea that moral terms are transparent is contested, though (cf. e.g. Sayre-McCord 1997).

5.3 The value of opaque terms in conceptual analysis

When a term is epistemically opaque, one cannot determine its secondary, i.e. metaphysical application conditions a priori. Does this mean that any kind of conceptual analysis is futile if it is concerned with such terms? As I argued mainly in chapter 3, opaque terms nevertheless have primary intensions. These a priori associations could serve as the basis for a conceptual analysis, even if they are insufficient to give us access to the associated secondary intensions. On the other hand, I also suggested, in chapter 3, to take the primary intension of the opaque term 'water' as comprising a theory about the natural kind water and proposed that at least one of its key functions is to secure reference to that natural kind. One might therefore suspect that, at least insofar as other opaque terms are similar, the primary intensions of such terms themselves are not particularly important.

But these are quite abstract considerations, and their implications for conceptual analysis are not obvious. Take an epistemically transparent expression such as 'the most massive organism in the world': This expression will pick out the most massive organism, whatever it is, in every world considered as actual and in every world considered as counterfactual. Rigidifying that expression yields 'the actually most massive organism in the world'. That expression is opaque; in order to determine its reference with respect to worlds considered as counterfactual, one needs to have empirical information about the character of the actual world. Nevertheless, it is hard to see why (the primary intension of) this latter expression should contain less information than (that of) the former one. On the face of it, anything which can be extracted from the unrigidified expression can also be extracted from the rigidified one. Similar considerations apply to other opaque expressions: Assume for example that natural kind terms like 'water' are, as was argued in chapter 3, rigidified theoretical terms. Why should such a term be less interesting for conceptual analysis than a corresponding term which picks out, say, the respective occupant of the water role in any metaphysically possible world, instead of rigidly picking

out the actual occupant of the water role? In each case, the same theoretical role can be carved out by an analysis of the term's primary intension. So prima facie, if one takes the aim of conceptual analysis to be an analysis of an expression's primary intension, then it need not matter whether that term is transparent or opaque.

Things are not quite that simple, however. Firstly, in many cases we are interested in a category's 'metaphysical essence' which is expressed by the secondary intension of the corresponding expression. And secondly, if the main function of the primary intensions of opaque terms is indeed to secure reference to the denoted category, this may suggest that they are typically not shared among speakers. Since both of these considerations put the value of the conceptual analysis of opaque expressions into doubt, I will address them in the following.

5.3.1 Discovering essences

Let me start with the former of the considerations just mentioned. If, for instance, one aims to reveal the nature of water, an analysis of the term 'water' alone is obviously insufficient. So the fact that 'water' is opaque at least diminishes the value of an analysis of the term. One could try to generalize this point. Suppose that one deals with a question of the form 'What is X?', and someone comes up with the answer 'X is the actual occupant of the X-role'. Even if the 'X-role' is spelt out in detail, the question will still at most be half-answered: We still need to know what it is that actually occupies the X-role. This suggests that what we are really interested in in philosophical inquiry and elsewhere is secondary intensions. And if so, then the utility of conceptual analysis involving opaque expressions seems questionable.

I think it is not so clear that we are always interested in the secondary intension associated with an expression or that we should always be interested in secondary intensions. At least I see no reason why a term's secondary application conditions are generally to be considered a more worthwhile target of inquiry than its primary application conditions.

Nevertheless, it cannot be denied that we can and often do try to reveal secondary intensions. In such cases, conceptual analysis alone cannot give us what we want if the relevant expression is epistemically opaque. This does not mean that conceptual analysis has no role to play in such an enterprise, which can rather be construed as a two-step process: In the first step, the primary intension of the relevant expression is revealed. The second step is to determine empirically what the primary intension picks out.[116] It is true that the first of these steps does not have to be made explicitly. Nevertheless, as I argued mainly in chapter 4, primary intensions are required to determine the secondary intensions even of opaque terms.

5.3.2 Variation in primary intensions

Now let me turn to the second potential source of concern mentioned above regarding the value of opaque expressions in conceptual analysis. I just suggested that it can be a worthwhile project to analyze the primary intensions even of opaque expressions. This presupposes that there is *a* primary intension which is associated with the relevant expression, i.e. one which is at least approximately shared among speakers. But if the main function of opaque expressions is to secure reference to the relevant category, this provides reasons to doubt that the primary intensions of opaque expressions are usually shared. In fact, Anti-Fregeans often point out that the conception (or whatever they call it) a subject associates with for example a natural kind term is irrelevant. Even a Martian for whom 'water' denotes a poisonous gas sometimes used as a chemical weapon

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[116] Compare for example Jackson's reconstruction of the identification of the temperature in gases with mean molecular energy:
"Pr. 1 Temperature in gases = that which plays the temperature ('T') role in gases. (Conceptual claim)
Pr. 2 That which plays the temperature role in gases = mean molecular kinetic energy. (Empirical discovery)
Conc. Temperature in gases = mean molecular kinetic energy. (Transitivity of '=')" (Jackson 1998a, 59)

could still possess the concept of water, as long as its ‘water’ utterances or thoughts referred to H_2O.
As we saw in 4.2, it is not obvious that it does not matter which primary intension a subject associates with an opaque term; there are at least some opaque terms where a specific primary intension has to be associated, most clearly in the case of indexical expressions.
Recall, moreover, that in 4.2, I also gave an argument based on the success of (communication in) mutual inquiries in support of the claim that the primary intensions of opaque expressions are usually shared among speakers. If that conclusion is correct, this is sufficient for an analysis of such expressions to be potentially useful. There is thus reason to think that most epistemically opaque expressions are suitable for a conceptual analysis.

Let me sum up the results of this chapter: I first showed that there are opaque as well as transparent expressions and identified ways to show on which side of this divide an expression falls. It transpired that semantic neutrality and epistemic transparency can come apart: While it is sometimes only empirically evaluable whether an expression is semantically neutral, epistemic transparency can be discovered a priori. Then I discussed the potential value of opaque terms in conceptual analysis. I conceded that in many cases where opaque terms are involved, conceptual analysis (alone) cannot give us everything we want. But I also noted that much depends on what one aims at. If one is interested in the ‘metaphysical essence’ of the category denoted by an opaque term, then conceptual analysis can only do part of the job. Nevertheless, an analysis of such a term – revealing its epistemic application conditions – can be of interest in its own rights. To this I should add that conceptual analysis need not be understood as a method which always results in an explicit (linguistic) analysis of a term. The Canberra Plan (which will be introduced in chapter 7), for example, which essentially involves conceptual analysis, can be applied to epistemically transparent and to opaque expressions equally well. And for conceptual analysis to play a role in reductive explanation,

the terms involved do not have to be epistemically transparent, either. I will discuss the potential aims and applications of conceptual analysis in detail in chapter 7.

Addendum: Does incomplete understanding present a problem for conceptual analysis?

At the beginning of this chapter, I conceded that firstly, deferential concepts are epistemically opaque and secondly, all kinds of expressions can be used deferentially, but argued that this does not entail that opacity is ubiquitous. The reason is that such terms can still be (and often are) transparent for those with complete understanding. What I left open, however, was whether the phenomenon of deferential/incomplete understanding itself presents a problem for conceptual analysis. At least at first glance, there are reasons for thinking so. For a start, the primary intension of a deferential concept does not seem to contain particularly useful information. Above I said that the analysis of opaque expressions can be worthwhile; but discovering something like 'X is the stuff called 'X' by the relevant experts' seems hardly of interest.[117] However, this observation only reveals a problem if people with an incomplete or deferential understanding of a term were to try and analyze it. One could thus give the simple advice only to analyze expressions which one understands completely or non-deferentially – which seems natural enough. There is a catch, however: As I already pointed out in chapter 3, while it is a priori whether a concept is deferential, incomplete understanding, and in particular, misunderstanding, may not be accessible to a speaker. There is therefore a risk that a person analyzes a term which she does not understand correctly. In such a case, the analysis may provide substantial information, but only about the subject's individual concept.

[117] I should note, however, that as was already mentioned in chapter 3, scrutability applies to deferential concepts as well.

In fact, this is just a special case of a problem which was discussed in a slightly different context at the end of chapter 3. There, I said that conceptual analysis has an a priori part and an empirical part. The a priori part consists in an analysis of a subject's individual concept, i.e. of the properties she associates with a given term. In the empirical part, one needs to check whether this concept is shared by other members of the linguistic community (and whether the associated properties can thus be said to reflect the term's linguistic meaning). That latter part is necessary to ensure that there is a common subject matter, since we can never know a priori whether a concept is shared by others. In this respect, incomplete understanding does not pose a special problem to conceptual analysis.

There is another problem which could be raised by deferential/incomplete understanding. There may be terms which are incompletely understood by the majority of those who use them. In fact, I think this is not a rare phenomenon. Just consider terms such as 'boson', 'transistor', 'microchip', 'vertebrate', etc. Now the problem is that I said above that for conceptual analysis to be (part of) a common enterprise, the concepts in question have to be shared among speakers of a linguistic community. But if there is only a small group of people – the 'experts' – with a complete understanding of the term, as is plausible in many cases, then this criterion seems to be violated. Or maybe rather, what is widely shared is the deferential understanding, not the complete understanding.

I think this problem is not all that worrisome. First of all, it can be doubted that such terms are especially interesting for a conceptual analysis anyway: Those terms which have been the target of conceptual analysis are typically everyday terms which are well understood by the average speaker. It would presumably make less sense, for example, to analyze technical terms. Goldman argues roughly along these lines for the claim that philosophically relevant terms are understood completely by the average speaker (cf. Goldman 2007, 18).

While I think there is something to this idea, I concede that it would be too bold to claim that we are never interested in an analysis of a term which is incompletely understood by many. So here is a second reason why I think

the present challenge to conceptual analysis is not too serious: When most of the speakers within a linguistic community defer to a group of experts, then they concede, at least implicitly, that the properties associated by these experts are the relevant ones. Therefore, in such cases it is the application conditions identified by an analysis of the experts' concept which are of interest. These might even be said to mirror the term's linguistic meaning: At least I see no principled reason why one should not say that only a relatively small group of speakers within a linguistic community has complete understanding of a term.

I conclude that the phenomenon of deferential or incomplete understanding does not provide any principled obstacles to conceptual analysis. The main reason is that it is a priori whether a concept is deferential; we can thus leave the analysis of concepts to those with non-deferential (and thus typically complete) understanding.